

KITS KKR & KSR INSTITUTE OF TECHNOLOGY AND SCIENCES (AUTONOMOUS)

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Broadcast Tree connections:

Subnet 1 connected to Subnet 2 (Cost = 2)

Subnet 2 connected to Subnet 3 (Cost = 3)

Subnet 1 connected to Subnet 4 (Cost = 6)

Total cost of broadcast tree = 11

EXPERIMENT-13

AIM : Wireshark

- i. Packet Capture Using Wire shark
- ii. Starting Wire shark using command line
- iii. Viewing Captured Traffic
- iv. Analysis and Statistics & Filters.

Wireshark is an open-source packet analyzer, which is used for education, analysis, software development, communication protocol development, and network troubleshooting.

It is used to track the packets so that each one is filtered to meet our specific needs. It is commonly called as a sniffer, network protocol analyzer, and network analyzer. It is also used by network security engineers to examine security problems.

Wireshark is a free to use application which is used to apprehend the data back and forth. It is often called as a free packet sniffer computer application. It puts the network card into an unselective mode, i.e., to accept all the packets which it receives.

A packet is a unit of data which is transmitted over a network between the origin and the destination. Network packets are small, i.e., maximum 1.5 Kilobytes for Ethernet packets and 64 Kilobytes for IP packets. The data packets in the Wireshark can be viewed online and can be analyzed offline.

Installation of Wireshark Software.

Below are the steps to install the Wireshark software on the computer:

- Open the web browser.
- Search for 'Download Wireshark.'
- Select the Windows installer according to your system configuration, either 32-bit or 64-bit. Save the program and close the browser.
- Now, open the software, and follow the install instruction by accepting the license.
- The Wireshark is ready for use.

On the network and Internet settings option, we can check the interface connected to our computer.

If you are Linux users, then you will find Wireshark in its package repositories.

i. Packet Capture Using Wireshark Capturing your traffic with Wireshark After starting Wireshark, do the following:

1. Select Capture | Interfaces.
2. Select the interface on which packets need to be captured. This will usually be the interface

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where the Packet/s column is constantly changing, which would indicate the presence of live traffic). If you have multiple network interface cards (i.e. LAN card and Wi-Fi adapter) you may need to check with your IT administrator to determine the right interface.

3. Click the Start button to start the capture.
4. Recreate the problem. The capture dialog should show the number of packets increasing. Try to avoid running any other internet applications while capturing, closing other browsers, Instant messengers etc
4. Once the problem which is to be analyzed has been reproduced, click on Stop. It may take a few seconds for Wire shark to display the packets captured.
5. Save the packet trace in the default format. Click on the File menu option and select Save As. By default Wire shark will save the packet trace in libpcap format. This is a filename with a .pcap extension.

ii. *Starting Wire shark using command line*

You can start Wire shark from the command line, but it can also be started from most Window managers as well. In this section we will look at starting it from the command line.

Wire shark supports a large number of command line parameters. To see what they are, simply enter the command *wireshark-h*.

Wire shark 3.7.3 (v3.7.3rc0-25-g72703582d587) Interactively dump and analyze network traffic.
See <https://www.wireshark.org> for more information. Usage: wireshark [options] ... [<infile>]

Capture interface:

- i <interface>, --interface <interface>
name or idx of interface (def: first non-loopback)
- f <capture filter> packet filter in libpcap filter syntax

iii. *Viewing Captured Traffic*

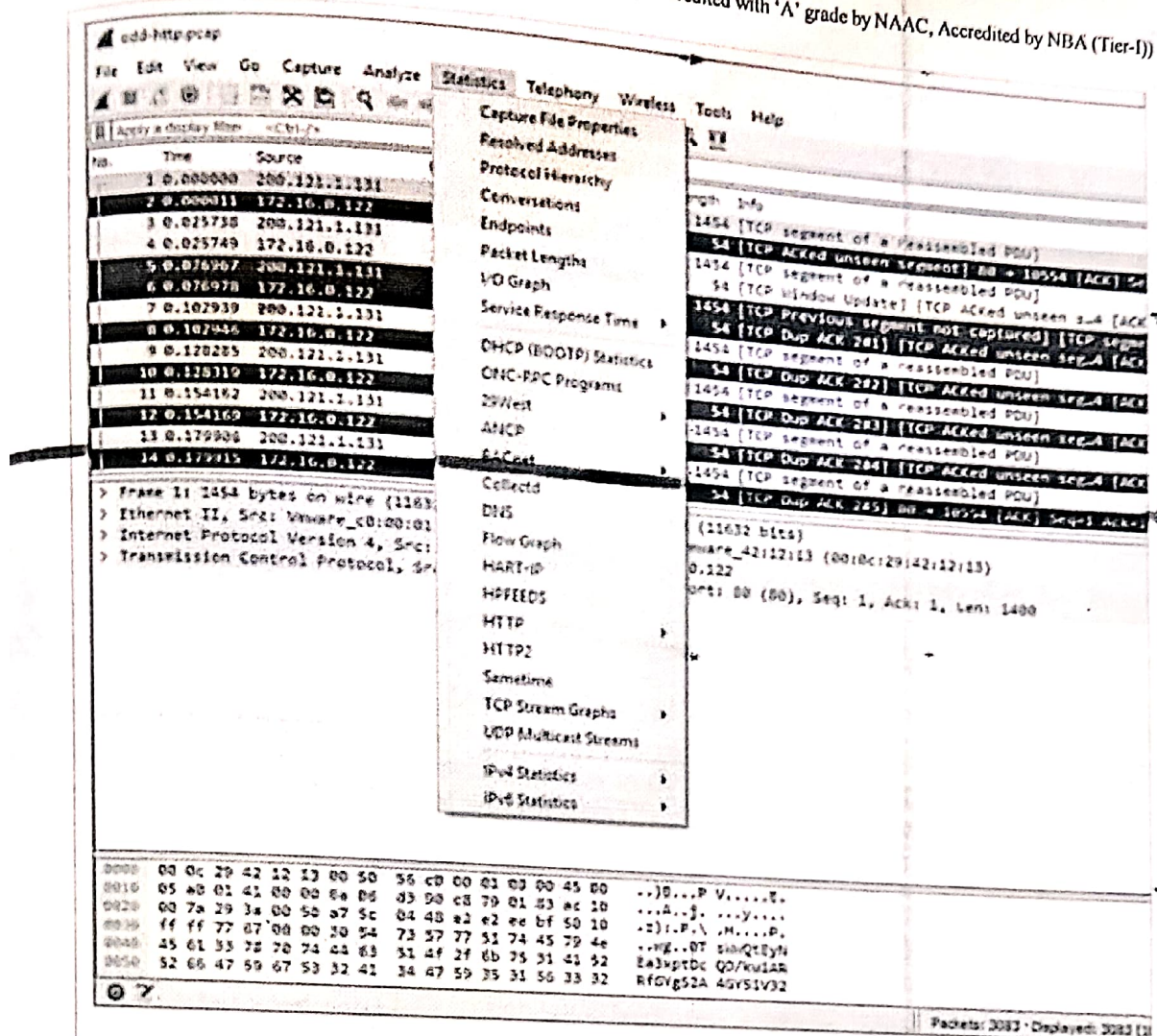
Once you have captured some packets or you have opened a previously saved capture file, you can view the packets that are displayed in the packet list pane by simply clicking on a packet in the packet list pane, which will bring up the selected packet in the tree view and byte view panes. Once you have captured some packets or you have opened a previously saved capture file, you can view the packets that are displayed in the packet list pane by simply clicking on a packet in the packet list pane, which will bring up the selected packet in the tree view and byte view panes.

iv. *Analysis and Statistics & Filters Analysis menu in wireshark*

The "Analyze" Menu The Wireshark Analyze menu contains the fields shown in Analyze menu items.

Filters in wireshark

Wireshark has two filtering languages: capture filters and display filters. Capture filters are used for filtering when capturing packets, "Filtering while capturing". Display filters are used for filtering which packets are displayed.



EXPERIMENT-14

AIM : How to run Nmap scan.

1. **Download the Nmap installer :** This can be found for free from the developer's website. It is highly recommended that you download directly from the developer to avoid any potential viruses or fake files. Downloading the Nmap installer includes Zenmap, the graphical interface for Nmap which makes it easy for newcomers to perform scans without having to learn command lines.
 - The Zenmap program is available for Windows, Linux, and Mac OS X. You can find the installation files for all operating systems on the Nmap website.
2. **Install Nmap :** Run the installer once it is finished downloading. You will be asked which components you would like to install. In order to get the full benefit of Nmap, keep all of these checked. Nmap will not install any adware or spyware.
3. **Run the "Nmap – Zenmap" GUI program :** If you left your settings at default during installation, you should be able to see an icon for it on your desktop. If not, look in your Start menu. Opening Zenmap will start the program.
4. **Enter in the target for your scan :** The Zenmap program makes scanning a fairly simple process. The first step to running a scan is choosing your target. You can enter a domain (example.com), an IP address (127.0.0.1), a network (192.168.1.0/24), or a combination of those.

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5. **Choose your Profile :** Profiles are preset groupings of modifiers that change what is scanned. The profiles allow you to quickly select different types of scans without having to type in the modifiers on the command line. Choose the profile that best fits your needs:[1]

- **Intense scan** - A comprehensive scan. Contains Operating System (OS) detection, version detection, script scanning, traceroute, and has aggressive scan timing. This is considered an intrusive scan.
- **Ping scan** - This scan simply detects if the targets are online, it does not scan any ports.
- **Quick scan** - This is quicker than a regular scan due to aggressive timing and only scanning select ports.
- **Regular scan** - This is the standard Nmap scan without any modifiers. It will return ping and return open ports on the target.

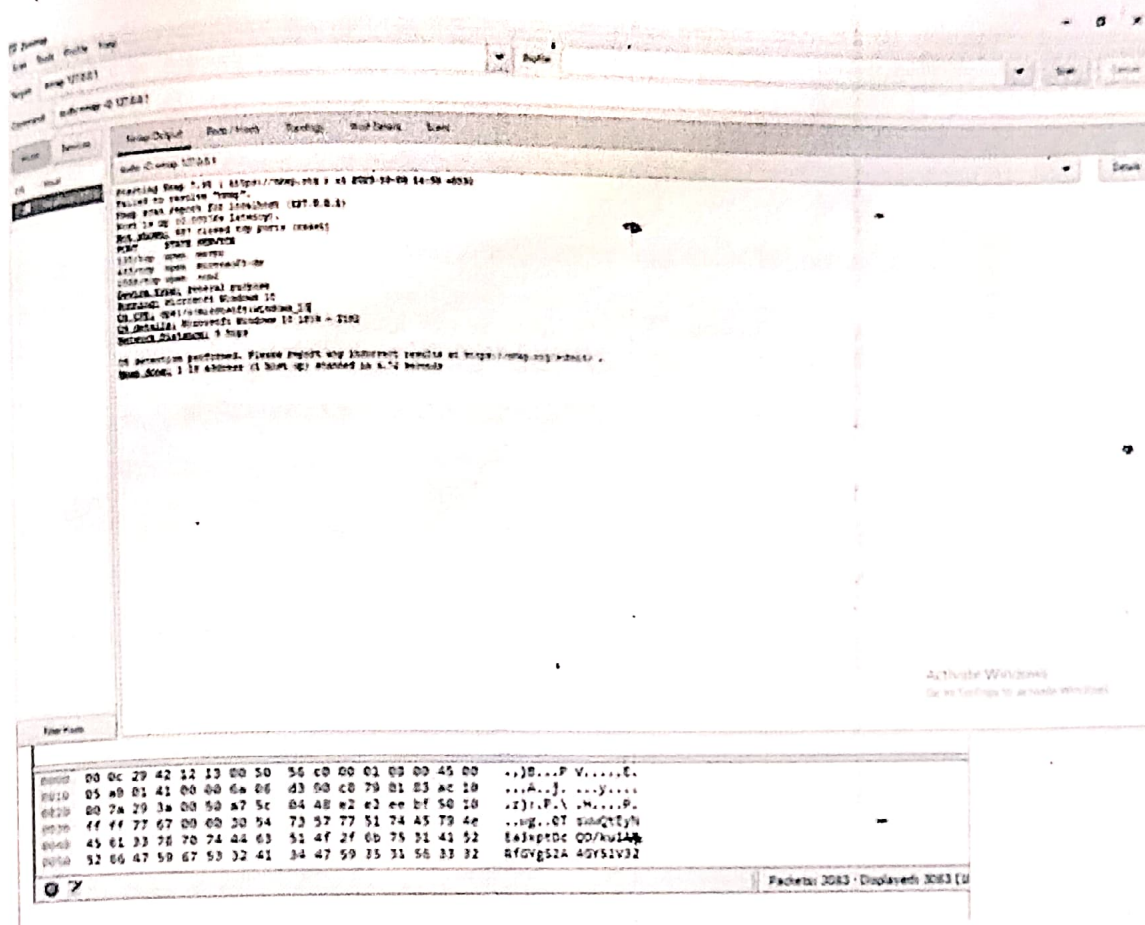
6. **Click Scan to start scanning:** The active results of the scan will be displayed in the Nmap Output tab. The time the scan takes will depend on the scan profile you chose, the physical distance to the target, and the target's network configuration.

7. **Read your results :** Once the scan is finished, you'll see the message "Nmap done" at the bottom of the Nmap Output tab. You can now check your results, depending on the type of scan you performed. All of the results will be listed in the main Nmap Output tab, but you can use the other tabs to get a better look at specific data.[2]

- **Ports/Hosts** - This tab will show the results of your port scan, including the services for those ports.
- **Topology** - This shows the traceroute for the scan you performed. You can see how many hops your data goes through to reach the target.
- **Host Details** - This shows a summary of your target learned through scans, such as the number of ports, IP addresses, hostnames, operating systems, and more
- **Scans** - This tab stores the commands of your previously-run scans. This allows you to quickly re-scan with a specific set of parameters.

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EXPERIMENT-15

AIM : Operating System Detection using Nmap

Program:

nmap -O -sV x.x.x.x

Starting Nmap 7.92 (<https://nmap.org>) at 2021-09-17 11:02

PDT Nmap scan report for x.x.x.x

Host is up (0.22s latency).

Not shown: 994 closed tcp ports (reset)

PORT STATE SERVICE VERSION

135/tcp open msrpc Microsoft Windows RPC

139/tcp open netbios-ssn Microsoft Windows netbios-ssn 445/tcp open microsoft-ds

2701/tcp open cmrccservice Microsoft Configuration Manager Remote Control service (CmRcService.exe)

3306/tcp open mysql MySQL 5.1.60-community-log 3389/tcp open ms-wbt-server Microsoft Terminal Services

Aggressive OS guesses: Microsoft Windows Server 2016 (93%),
 Microsoft Windows Server 2008 R2 SP1 (92%), Microsoft Windows Server 2012 R2 (91%),
 Microsoft Windows 10 1607 (90%), Microsoft Windows Server 2008 R2 (90%), Microsoft
 Windows Server 2012 (89%), Microsoft Windows Server 2012 or

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Windows Server 2012 R2 (89%), Microsoft Windows Server 2008 R2 or Windows 8 (87%),
Microsoft Windows Server 2008 R2 SP1 or Windows 8 (87%), Microsoft Windows 7 SP1 (86%)
OS matches for host (test conditions non-ideal).

Network Distance: 18 hops

Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

<https://nmap.org/sublinux/>.
Nmap done: 1 IP address (1 host up) scanned in 33.77 seconds

